JOEL ANTHONY

Electrical Engineer

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Electrical engineer with hands-on experience in STM32 motor-controller firmware, PCB design, CAN and test automation, now owning sensor and switch components for transmission controls. Building an STM32 + FPGA platform for robotics controls on my own time, and starting Purdue's MSECE pathway in Spring 2027.

EXPERIENCE

Allison Transmission — Controls Systems Electro-Mechanical Component Engineer — Jun 2024 – Present

- Primary responsible engineer for pressure sensors and pressure switches across conventional, hybrid and electrified propulsion systems.
- Led a product-line transition from pressure switches to pressure sensors.
- Wrote specifications and statements of work for the supplier RFQ on new pressure sensors, and design notebook documentation for pressure switch changes.
- Identify component requirements with internal and external customers; test and validate hardware.
- Perform root-cause analysis on embedded control hardware failures, review electrical design decisions in PSpice and OrCAD, and work with suppliers on manufacturing process improvements.

Hydro-Gear — Electrical Engineer *(Product Engineering Intern Apr 2022 – May 2023)* — Apr 2022 – Jun 2024

- Wrote and debugged STM32G4 firmware for electric motor controllers in Keil µVision and STM32CubeIDE, including six-step motor control.
- Debugged and implemented CAN communication using LabVIEW with NI-XNET, Vehicle Spy and ValueCAN.
- Streamlined and automated motor efficiency testing on Magtrol dynamometer equipment; implemented NI hardware and software for end-of-line testing.
- Designed schematics and PCBs in Altium, set up Altium version control, prepared contract-manufacturer documentation and root-caused failed boards; ran DFMEAs.

Applied Intelligence Corporation — Electrical Engineering Intern — Aug 2021 – Mar 2022

- Built and wired control panels and custom wiring harnesses from customer schematics; produced CNC-engraved labels.

BorgWarner / Delphi Technologies — Manufacturing and Test Engineering Intern — May 2019 – Aug 2021

- LabVIEW data acquisition for the manufacturing test group; debugged test equipment and presented findings to management.

King & Anthony Lawn Care, LLC — Co-Founder — Apr 2017 – Apr 2021

- Started and ran a lawn care business with a seasonal crew, from client acquisition to equipment and year-end financials.

SELECTED PROJECTS

STM32 + FPGA control platform — SPI register link, Nucleo-G431KB ↔ Basys 3 — 2026 – Present

- CRC-checked SPI register protocol with burst transfers, loop-tick and interrupt lines; Verilog RTL, C driver, Icarus testbench and driver-replay test in CI.

STM32 temperature and humidity monitor — custom Altium PCB, two revisions — Mar – Aug 2023

- STM32G431 board with BME280, I2C LCD and ESP8266 uploading to ThingSpeak; Nucleo prototype, then Rev A and B.

CORDIC engine for Tiny Tapeout — sky130 ASIC tile — 2026 – Present

- 16-bit fixed-point dual-mode CORDIC with a Python golden model; passes Tiny Tapeout GDS, precheck and gate-level test.

EDUCATION

Purdue University — B.E. Electrical and Electronics Engineering, 2023 · MSECE pathway from Spring 2027 — 2019 – 2023

- Senior design: developed a CAN module for Hydro-Gear's vehicles, working closely with Hydro-Gear engineers.

SKILLS

EMBEDDED	C, STM32 (HAL, STM32CubeIDE, Keil µVision), CAN, UART/SPI/I2C, six-step motor control
HARDWARE	Altium Designer, KiCad, OrCAD Capture CIS, PSpice, Fusion 360, PCB bring-up and debug
FPGA / RTL	Verilog, Xilinx Vivado, Icarus Verilog, Basys 3 and Arty A7
TEST	LabVIEW, NI-XNET, Vehicle Spy / ValueCAN, Magtrol dynamometers, end-of-line test
PROCESS	DFMEA, PPAP, RFQ specifications, root-cause analysis, supplier development
SOFTWARE	Python, Git (GitHub, Bitbucket), Linux / Ubuntu, Raspberry Pi